01 / TECHNICAL DOCUMENTATION

SME finance, with bounded AI decisions.

BizWizard | 2026 FinTechathon International Track
Topic E - SME Finance Copilot

Purpose. Help a small-business owner compare financing, anticipate cash pressure and understand business performance. Financial calculations remain deterministic; AI selects or explains constrained options.

What the prototype does

Topic E capability	Implemented workflow
Financing-option matching	Build a shop profile; apply eligibility rules; rank feasible products; price and negotiate simulated offers; create an owner-approved application and repayment schedule.
Cash-flow management	Maintain cash events and commitments; project a 90-day balance; raise rule-based alerts; propose guarded treasury actions and link recorded collection evidence.
Business financial analysis	Compute product economics and working-capital metrics; verify candidate explanations; compare scenarios; measure 30-day observations and matched sandbox outcomes.

How this document relates to scoring

Task completion (40%): the three workflows and their execution evidence. **Security & compliance (30%):** authorization, financial guards and the separate self-assessment. **Innovation & interaction (30%):** bounded agent negotiation, editable mandates, explanations and reproducible paired evaluations. These are evidence mappings, not predicted marks.

Scope and release

The application is a sandbox finance copilot, not a lender, payments processor or regulated financial adviser. No external lender is contacted and no bank transfer is executed. The project plan is design intent; this document describes the implementation and its measured limits.

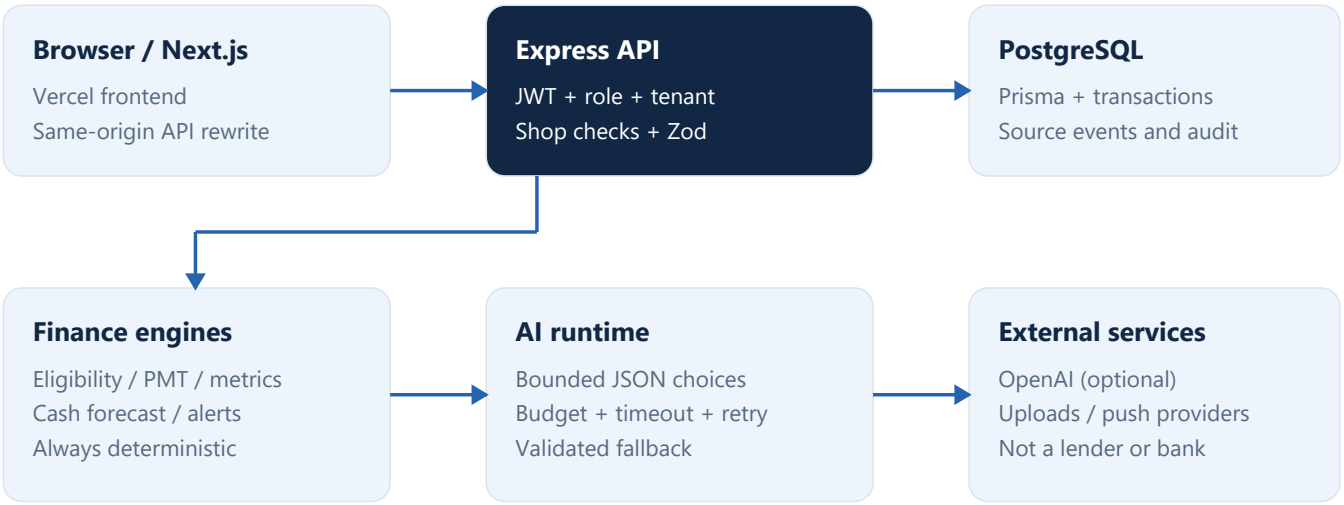
Evidence baseline: commit 8f394d2d074b2eafa93c4f7d283550aeb96f21dd. Six reports were generated from a clean tree on 18 October 2026 and then copied to the submission evidence folder. Live URL: https://bizwizard.sadman.app (project-provided; latest deployment not independently certified here).

Source anchors. Competition appendix, pp. 2-3; README.md; PLAN.md (design intent).

02 / SYSTEM ARCHITECTURE

One application database. Explicit trust boundaries.

A Next.js frontend, an Express backend and PostgreSQL, with optional external AI.



Trust boundaries: browser | application/VPS | database | optional external providers

Security design and data flow

The browser sends authenticated requests to the frontend's same-origin /api/proxy/v1 rewrite. Next.js forwards to the configured backend. Express verifies the token and required role, checks tenant membership and employee shop assignment, validates the request, then invokes a scoped service. CORS is not the authorization boundary.

Layer	Responsibility
Presentation	Next.js / React / Tailwind; dashboard workflows, light/dark themes, loading states and owner/employee controls. Hidden buttons are convenience, not access control.
Finance service	Matching, cash flow, analysis, agent ledger and simulator modules. Pure TypeScript engines calculate money and constraints; service transactions persist approved state.
Persistence	One production PostgreSQL database managed through Prisma migrations. Sales, debts, lots and expenses feed finance records; policy versions, decisions, runs and evaluation ledgers support traceability.

The disposable local simulator_test database is test infrastructure, not a second required production database. In-app demo workspaces are separate, scoped shops copied from a source shop; a virtual clock isolates simulated dates. Source history must remain unchanged through copy, run and reset.

Key record families: FinancingOffer / Application / LoanSchedule; CashAccount / Transaction / Commitment / Alert / TreasuryAction; Insight / Scenario / CapitalDecision; AiAgentPolicy / AiDecision / AiRun / FinanceEvaluation. This is not full double-entry or bank-reconciled accounting.

Source anchors. client/next.config.ts; server/src/app.ts; finance/shared/database.ts; finance/simulator/workspace.ts; server/prisma/schema.prisma. Finance paths are relative to server/src/app/modules/.

03 / CORE ALGORITHM - FINANCING

Rules first. Negotiation within bounds.

A simulated financing market with a lender role and an owner-side borrower role.

Stage	Deterministic contract
1. Profile and completeness	Aggregate the shop history: revenue, COGS, operating expense, receivables, inventory, debt burden and cash. Matching requires at least 60% coverage of its 90-day sales window.
2. Eligibility gate	Check amount, tenor, business age, revenue, debt/revenue, sector and required-document flags. NEAR_MISS means exactly one soft threshold missed by at most 20%; it is not automatic approval.
3. Rank feasible products	Weighted score: affordability 30%, cost 25%, liquidity 20%, collateral 15%, flexibility 10%. AI enrichment is bounded to +/-15 score points; the blend is 60% deterministic and 40% bounded AI.
4. Price and negotiate	Pricing and risk arithmetic run in code. The roles may accept, counter or reject within a maximum of three rounds. Product limits, owner rate ceilings, exposure and affordability remain authoritative.
5. Owner confirmation	Acceptance creates the scoped application and rounded amortization schedule transactionally. Duplicate acceptance reuses the application. Submission is not disbursement.

Money and affordability

```
Monthly FCF = revenue - COGS - operating expense - debt service

Screening monthly EMI cap = max(0, FCF) x 0.50

Borrower installment feasibility: monthly payment / FCF <= 0.35

PMT = principal x r / (1 - (1 + r)^(-n)); r = annual rate (%) / 1200
```

The 50% screening capacity and 35% borrower feasibility ceiling are separate checks, not interchangeable promises. Product/owner bounds still apply. Currency boundaries round to two decimals, with the last schedule installment absorbing residual principal.

Model limitations. Credit default risk is a bounded heuristic, not a validated credit bureau score. The helper named effectiveApr uses annualized total cost / principal, not an IRR-based regulatory APR. Displayed quotes are simulated and should not be used as real credit commitments.

Source anchors. finance/shared/eligibility.ts; affordability.ts; guard.ts; money.ts; finance/agents/agentCore.ts; finance/matching/matching.service.ts; finance-hardening.integration.test.ts.

04 / CORE ALGORITHMS - CASH & ANALYSIS

From recorded events to actionable context.

Projection and financial analysis use code-computed values rather than model-generated arithmetic.

Cash synchronization and projection

PostgreSQL triggers synchronize paid sales, debt payments and expense events with cash records, including updates and deletions. Repeated backfill is idempotent and preserves manual entries. Explicit supplier obligations are commitments; settlement links an existing matching purchase instead of issuing another payment.

The projection builds daily historical inflow/outflow series, estimates recent levels and day-of-week seasonality, applies a clipped trend, and adds scheduled commitments and probability-weighted receivables. It rolls balances forward for 90 days and identifies the trough, first shortfall and buffer days.

Nine alert conditions	Behavior
Shortfall within 14 days; buffer below target	Prioritize potential liquidity pressure; low-history rules suppress inappropriate confidence.
Commitment due soon; debt older than 30 / 60 days	Show dated obligations and aging receivables. The 30- and 60-day checks are separate conditions of the overdue alert family.
Idle cash; revenue decline; expense spike; insufficient data	Flag idle balances, a 7-day revenue decline relative to the prior average, expenses above twice their category median, or fewer than 14 ledger days.

Analysis and hypothesis verification

The analysis module computes margins, product contribution, working capital, breakeven and concentration from scoped records. Candidate explanations are checked by removing the candidate contribution from the original gap. A claim is VERIFIED only if the nonzero gap's residual is at most 10%; this is accounting attribution, not causal discovery.

Scenario levers change price, collections, operating cost or product mix through an explicit simplified model. Approved capital decisions freeze an observation window; 30-day before/after profit observations exclude future rows and remain non-causal. Treasury collection evidence links unique existing payments and is gross cash, not incremental profit.

Assumptions: collection probabilities and scenario elasticity are heuristics. Receivable age can stand in for overdue age. Scheduled commitments can overlap ordinary historical expenses. Cash accounts are not externally reconciled; historical supplier obligations must be recorded explicitly.

Source anchors. finance/shared/cashEngine.ts; alerts.ts; expenseSpike.ts; verifier.ts; scenario.ts; finance/cashflow/cashflow.service.ts; finance/analysis/analysis.service.ts; cash-event migrations.

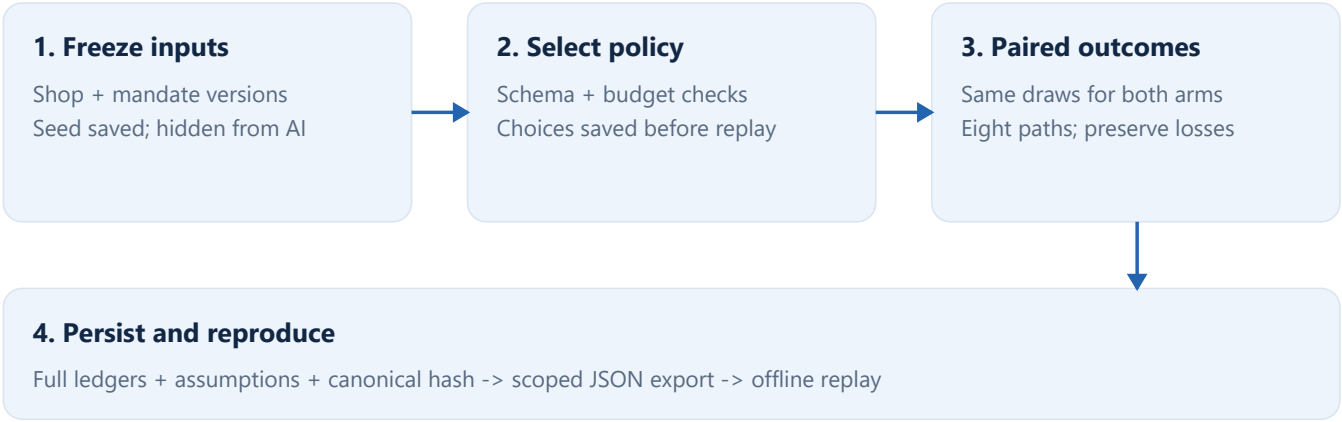
05 / AI RUNTIME & EVALUATION

A decision is not an observed outcome.

Separate model choices, deterministic calculations and measured evidence.

Each configured AI call requests JSON and validates it with Zod. The runtime permits one repair retry within its deadline. Missing keys, invalid output, budget denial and timeouts use a labelled deterministic fallback. Application calls reserve a conservative per-tenant monthly cost allowance and log model, prompt version, tokens, cost and status.

Shop-specific paired evaluation



The snapshot requires 60% sales-history completeness and at least 14 cash-history days. It captures current quotes, economics, commitments and mandate versions under a repeatable-read transaction. The model cannot choose seeds or future draws. A unique snapshot key and atomic run claim prevent concurrent duplicate paid calls; unchanged daily inputs reuse the saved evaluation.

Role	Paired sandbox metric
Lender	Full-term net credit cash return after funding costs, repayments and modeled recovery; common default draws. The pinned flat-rate baseline funds the same exposure-bounded universe.
Borrower	Full-term borrowing cost on equal-principal feasible quotes; baseline selects lowest quoted APR. Both sides assume repayment in full, independently of the lender experiment.
Treasurer / analyst	30-day closing cash versus delayed-collection rules; modeled capital gross margin versus equal allocation among revenue-ranked SKUs. The roles are independent experiments.

Evidence boundary. Eight paired paths are not a statistically established treatment effect. Quotes use captured current terms, not reconstructed pre-negotiation contracts. Missing data stays pending; losses and ties remain. Real-world alpha is not populated from sandbox results.

Source anchors. server/src/lib/ai/client.ts and budget.ts; finance/agents/matchedEvaluation.service.ts; matchedReplay.ts; benchmarks.ts; agents.service.ts.

06 / VERIFIED RESULTS

Mixed results. Reproducible evidence.

Measurements from clean source 8f394d2, generated on 18 October 2026.

Live AI versus pinned rules: three synthetic risk cases

Role	AI / rules / tie	Observed difference in favour of AI
Lender	1 / 2 / 0	BDT -4,000.00, -1,000.00, +5,000.00
Borrower	3 / 0 / 0	BDT +4,099.31, +4,099.31, +4,099.31
Treasurer	0 / 0 / 3	BDT +0.00, +0.00, +0.00
Analyst	0 / 3 / 0	BDT -60.00, -60.00, -60.00

Across 12 role/case comparisons: **4 AI wins, 5 rules wins and 3 ties**. Borrower cost fell by BDT 4,099.31 in each case; analyst modeled returns were BDT 60 lower. All three provider calls succeeded without fallback using gpt-4o-mini; total estimated API cost was USD 0.000273. Counts are not a combined financial return.

Separate shop-specific live sandbox run

Role	Baseline (BDT)	Participant (BDT)	Difference
Lender	665.94	265.22	-400.72
Borrower	705.82	423.28	+282.54
Treasurer	1,484,037.23	1,484,037.23	+0.00
Analyst	46,761.23	46,761.23	+0.00

These are means across eight paired paths: lender and borrower use their loan terms; treasurer and analyst use 30 days. The live policy succeeded without fallback; its estimated API cost was USD 0.000155. The exported result reproduced with a matching canonical outcome hash.

Database execution evidence

11 tests passed, zero failed or skipped across five suites, plus successful migration application. Coverage includes transactional cash events, supplier evidence, inventory/alert behavior, finance authorization and budgets, matching/application concurrency, isolated demo copy/run/reset, and frozen paired evaluations.

The standalone benchmark is not a replay of the full multi-agent negotiation pipeline. Fixture mode is not AI. Integration evidence is backend service/loopback HTTP evidence, not a browser video, production penetration test or real-world financial result.

Source anchors. evidence/submission/benchmark-live-results.json; matched-evaluation-live.json; matched-replay-verification.json; sandbox-operation-logs.json.

07 / REPRODUCTION & REVIEW

A short route from source to evidence.

The root README is the operational index. No production credentials are needed for offline checks.

Two runners, with different dependencies

```
# From server/ after npm ci

npm run benchmark:finance

npm run benchmark:finance -- --live

npm run evidence:finance

npm run evidence:finance -- --replay EXPORTED_EVALUATION.json
```

The benchmark needs Node.js 24, npm and Git; offline mode needs no database. Explicit --live uses the reviewer's own key in Git-ignored server/.env.benchmark or the environment. The execution runner additionally needs disposable local PostgreSQL database simulator_test at 127.0.0.1, configured through EVIDENCE_DATABASE_URL. It creates a new schema, applies migrations and disables external AI. Replay needs only the exported JSON and installed dependencies.

Application deployment

Frontend: Vercel, root client, Next.js preset, NEXT_PUBLIC_API_BASE_URL set to the backend /api/v1 endpoint. Backend: the original repository's main workflow builds an artifact and uses restricted SSH. The installed deployment script installs dependencies, generates Prisma, starts the backup service, applies committed migrations, switches the release and checks readiness. The VPS and its secrets/services must already be configured.

Cash triggers, supplier-obligation metadata and matched-evaluation storage are committed migrations. Use migrate deploy, not schema-only db push. Existing history can be reconciled using the owner-only backfill action. Code rollback does not reverse a database migration. A frontend fork must be synchronized separately.

Suggested reviewer path

Use a synthetic account or dedicated demo workspace. Open Cash Flow and inspect a projection and alert; run matching and inspect eligibility, negotiation and an approved application; inspect verified analysis; then run a matched evaluation in AI Agent Ledger, download it and replay the JSON. Review the separate Security Self-Assessment for permission tiers and residual risks.

Submission boundary: these two PDFs do not replace source access, the README, presentation or execution logs. The rules label the demo video optional in item 03 but also name it under non-on-chain evidence in item 06; supply a short video or confirm that interpretation with the organizer.

Source identity: 8f394d2d074b2eafa93c4f7d283550aeb96f21dd
Source-tree SHA-256: b9e886ce8b472a8532bfd782dada27028afb69550725452baba799e3cbe355fd
Repository: https://github.com/mahfuz-saim/bizwizard (private access requires invitation).

Source anchors. README.md; .github/workflows/backend.yml; server/deploy/deploy.py; server/scripts/benchmark-finance.ts; evidence-scenarios.ts; competition appendix, pp. 2-4.